

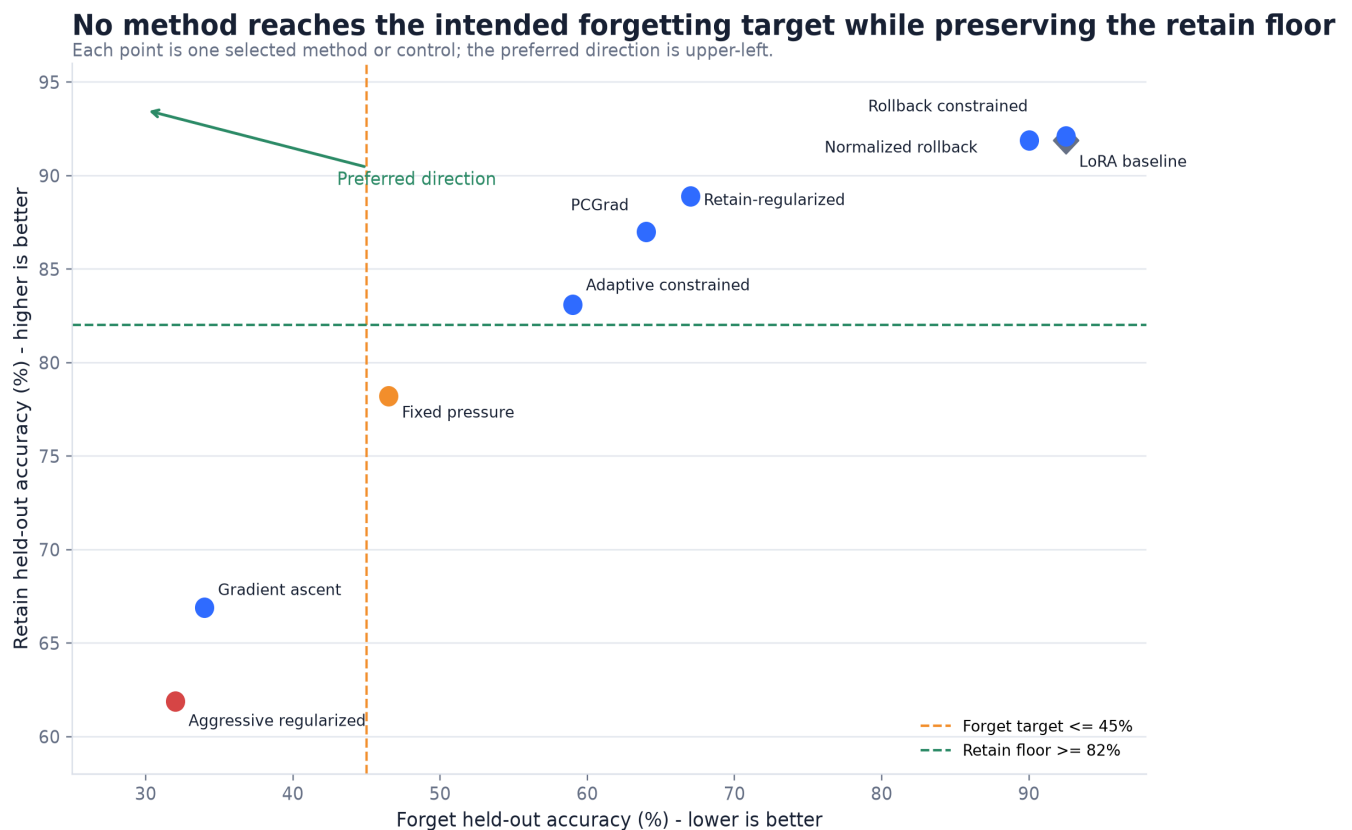
Machine Unlearning in a Small Language Model

Progress report: selective forgetting versus retained utility

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Executive summary

- The evaluated methods trace a clear frontier: stronger forgetting is consistently associated with lower retained-fact accuracy.
- No selected method simultaneously reaches the 45% forget held-out target and the 82% retain held-out floor.
- General-control accuracy falls from 88% in the base model to 56% after LoRA; with only 50 control questions, later 2-4 point differences are one or two answers.
- Rollback variants preserve utility but produce little measurable forgetting, making them useful negative results rather than successful unlearning methods.

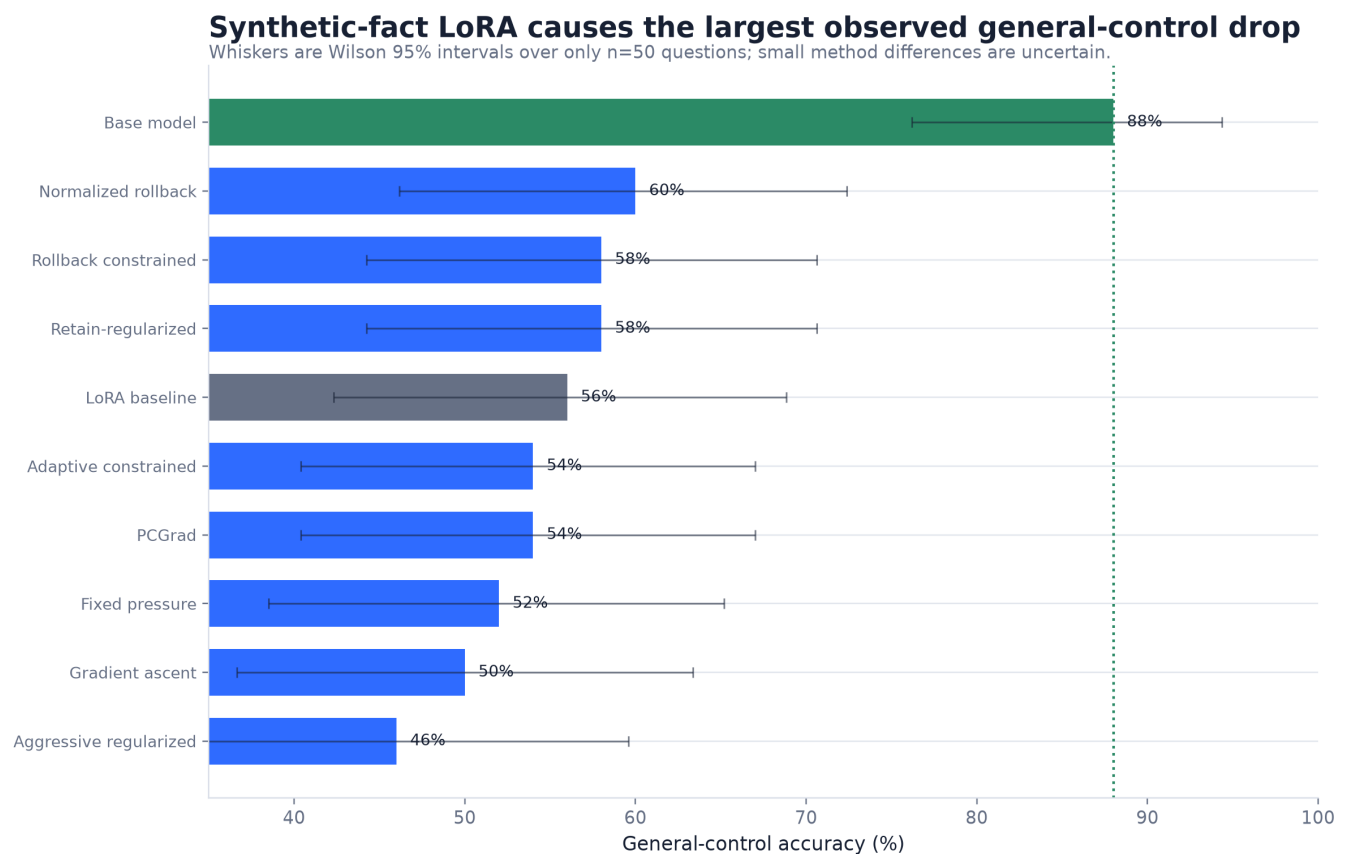


Experimental setup

Model	Qwen2.5-0.5B-Instruct with LoRA
Synthetic personas	100 total: 20 forget / 80 retain
Fact categories	5
Synthetic evaluation	1500 questions: 500 seen / 1000 held-out
Held-out sample sizes	forget n=200; retain n=800
General controls	n=50

Result uncertainty

The general-control set has only 50 questions. A 2-point difference is one answer, so Wilson intervals are shown in the chart and small method differences should not be ranked as meaningful.



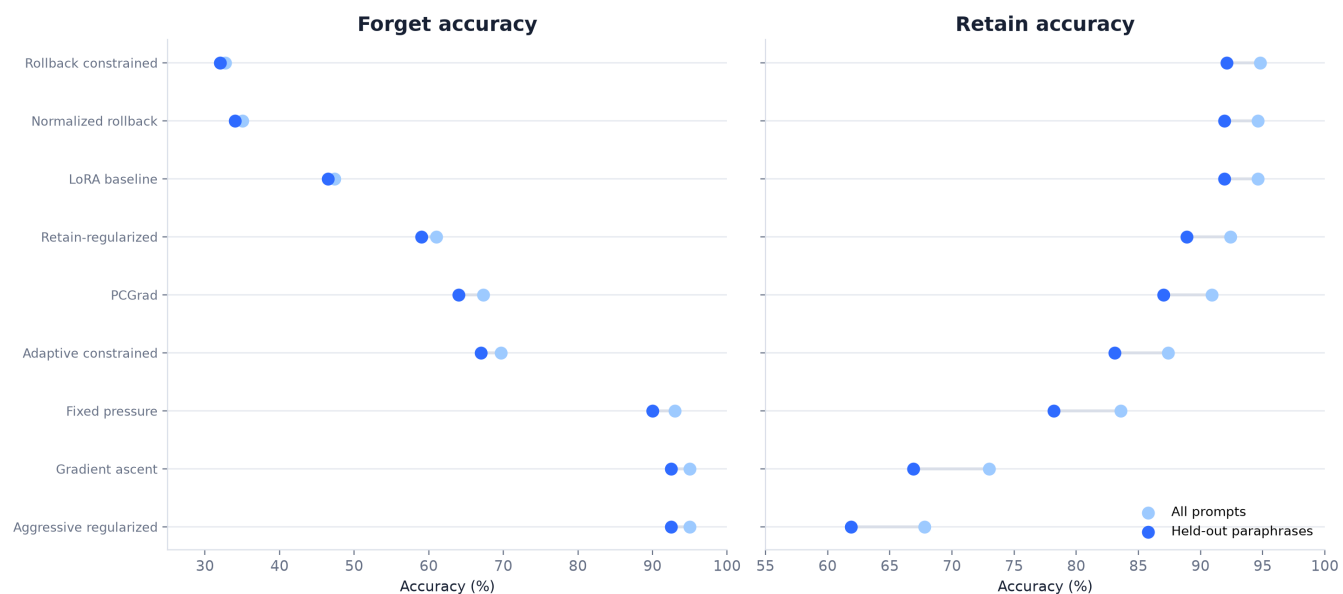
Complete final results

Method	Forget all	Forget held-out	Retain all	Retain held-out	General
LoRA baseline	95.0	92.5	94.6	91.9	56.0
Gradient ascent	35.0	34.0	73.0	66.9	50.0
Retain-regularized	69.7	67.0	92.4	88.9	58.0
Aggressive regularized	32.7	32.0	67.8	61.9	46.0
PCGrad	67.3	64.0	90.9	87.0	54.0
Adaptive constrained	61.0	59.0	87.4	83.1	54.0
Fixed pressure	47.3	46.5	83.6	78.2	52.0
Rollback constrained	95.0	92.5	94.8	92.1	58.0
Normalized rollback	93.0	90.0	94.6	91.9	60.0

Lower forget accuracy is better; higher retain and general-control accuracy are better.

Held-out paraphrases largely confirm the all-prompt ordering

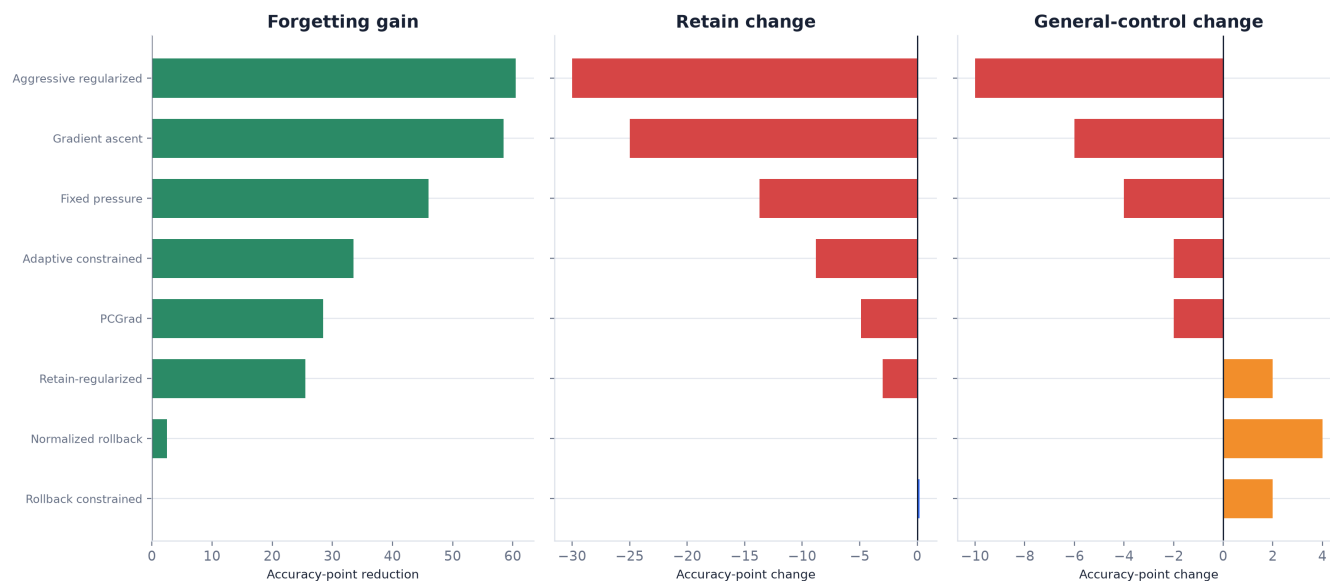
Larger gaps indicate weaker generalization across prompt wording.



Interpretation

Every meaningful forgetting gain is paid for with retained-knowledge loss

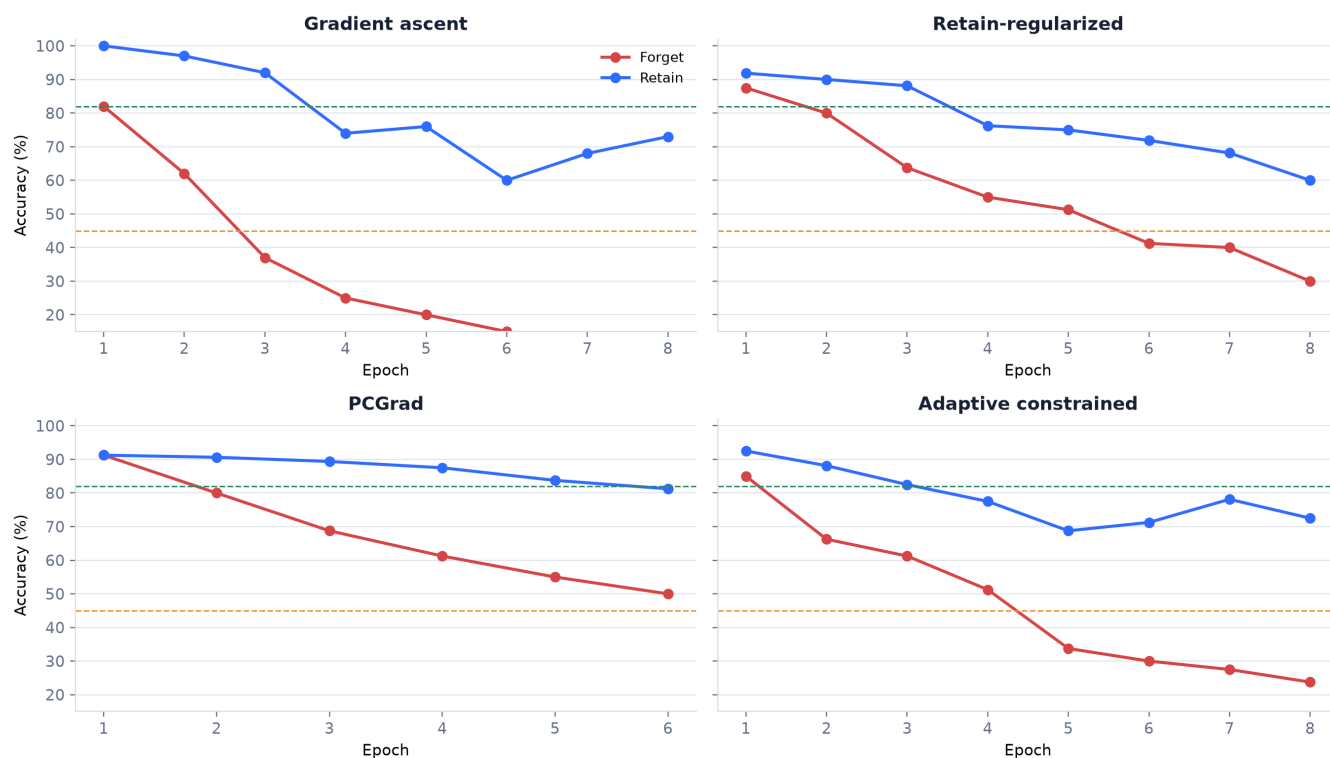
All changes are relative to the learned LoRA baseline.



The evidence supports a continuum rather than a universally best method. Adaptive is the strongest selected full-evaluation result that remains above the 82% retain floor, but it still misses the 45% forget target. Rollback variants are preservation-first negative results.

Training trajectories show where forgetting begins to damage retention

Week 4 uses train/sample accuracy; later weeks use held-out selection subsets.



Recommended next experiments

- Train the oracle baseline from the original base model using only retain examples.
- Repeat key methods across multiple seeds and report paired question-level uncertainty.
- Add extraction-oriented prompts, token probabilities, partial matches, and repeated sampling.
- Audit and reduce the 88% to 56% general-control drop caused by synthetic-fact LoRA.
- Only then replicate the stable protocol with Qwen2.5-1.5B.

Questions for the professor

- Which retain threshold should define an acceptable unlearning result?
- Should the next compute budget prioritize the oracle and multiple seeds before model scaling?
- Which evidence is required to support a claim of information removal rather than prompt-specific suppression?